

The cerebellum

The **cerebellum**, sometimes called the hindbrain, is at the back of the head, Fig. 1, and has a very distinctive structure which is largely preserved across species. It contains the numerous cells of the brain, the cerebellar granule cell and one of the largest, the Purkinje cell. In order to decide the degree to which different functions are preserved of different brain areas, the French neuroanatomist Jean Pierre Flourens performed a series of experiments in the early part of the nineteenth century in which he removed the cerebellum from living animals. He discovered that without a cerebellum animals were still able to move, but without their accustomed grace. This is consistent with what is observed for patients with cerebellar damage, see Fig. 2.

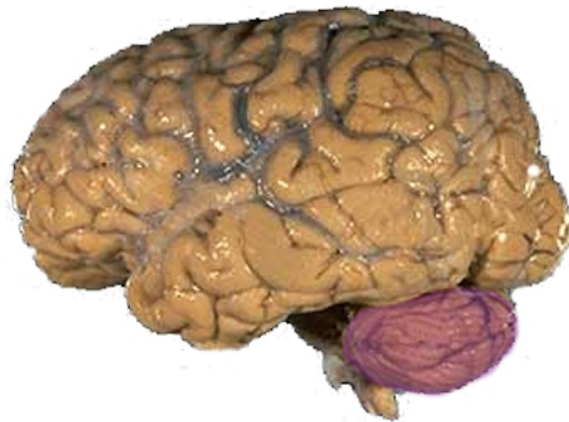


Figure 1: **The cerebellum**; a brain with the cerebellum marked in purple. Figure from the NIH

These days it is thought that the cerebellum performs very precise predictions in aid of movement control. A challenge in designing a control system is that the motor controls need to be based on a current estimate of position even though that estimate may not be available straight away because of the latency in sensory processing. A potential solution is to use a **forward model** to predict the current position from the available, past, sensory data and one common idea is that this is the role of the cerebellum.



Figure 2: **A woman with cerebellar damage**; this famous drawing from 1912 shows the distinctive gait associated with cerebellar damage, the patient is able to walk, but her walking seems self-conscious and awkward. Figure from Wikipedia.

The cerebellum as a perceptron

The **cerebellum** has a number of striking features; it has a more stereotypical structure than most brain area and this structure is conserved across species. It also has one of the brain's largest cells, the Purkinje cell, and its most numerous, the granule cell.

Purkinje cells have a distinctive structure with a huge, highly branched, but flat dendritic arbour, see Fig. 3; this allows an extensive connectivity with each Purkinje cell receiving inputs from around 100,000 other cells. In the cerebellum the Purkinje cells are lined up like pages in a book, with their arbours lying in parallel planes. They receive two excitatory inputs, weak inputs from parallel fibres, axons that run perpendicular to the planes of the Purkinje cell dendritic arbours, and a strong input from a climbing fibre, a single axon which winds around the Purkinje cell and makes multiple contacts with it, see Fig. 4.

Another peculiarity is that the Purkinje cell emits dramatically different types of spikes to different inputs. In response to multiple weak inputs from the parallel fibres it fires a normal spike, called in this context a **simple spike**. In response to single spike from the **climbing fibre** (from the **inferior oli-**



Figure 3: A drawing by Santiago Ramón y Cajal of a Purkinje cell. [Picture taken from http://en.wikipedia.org/wiki/Golgi's_method]

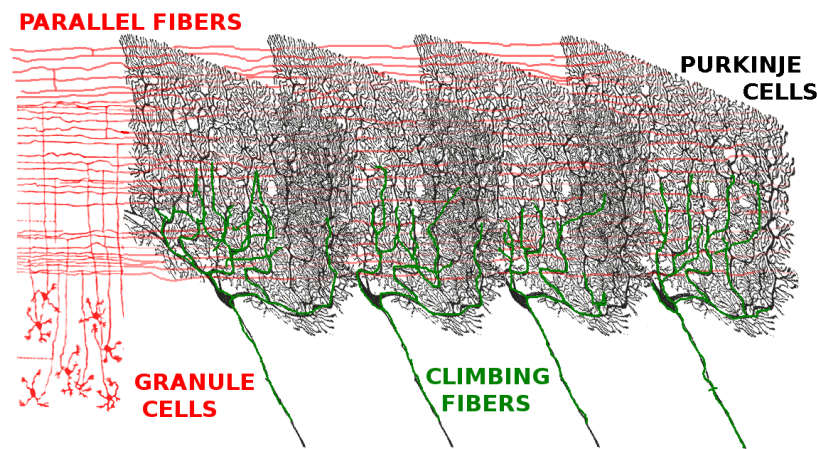


Figure 4: A cartoon of the cerebellar circuitry. A vertical axon rises from each granule cells, splits once and then extends horizontally in two directions making connections with multiple Purkinje cells. Each Purkinje cell has its own **climbing fiber** which winds up around it.

various nucleus) is fires a special spike, called a **complex spike**, with a leading spike, a number of small 'spikelets' and a sustained after-period of depolarisation; this is illustrated in Fig. 5. Complex spikes are believed to carry an *error signal*, to trigger *long-term depression* in recently active granule → Purkinje cell synapses.

It is still unclear exactly what the cerebellum does; what is known is that it is important for actions, fine motor control and proprioception; problems with the cerebellum are associated with ataxia, loss of fine motor control, poor motor learning and poor balance. There is a specific gait associated with cerebellar damage, one that exhibits a certain self-consciousness or vigilance is required for movement. According to most ideas about cerebellar function it is required for the calculation of fine motor signals [1], or for predicting the sensory or proprioceptive consequences of motor actions [2]; it is believed it encodes forward and backwards models of movement so that it predicts the consequences of motor commands, or calculates the motor command that will a particular movement.

Whatever exactly it does, it is widely believed, in accordance with the Marr-Albus model [3, 1], that the connections from parallel fibers to Purkinje cells

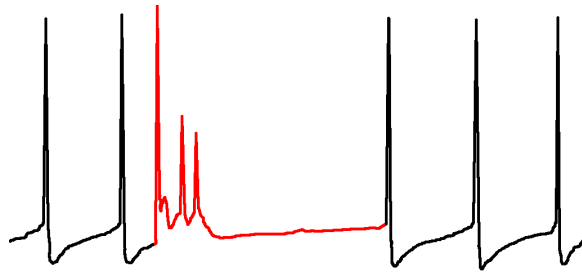


Figure 5: A complex spike. This drawing shows simple spikes in black and a complex spike in red. The complex spike is followed by a long refractory period during which spiking is not possible. This is a sketch, not an actual recording, but a typical time scale would have this refractory period 50 ms long.

acts as a perceptron. Thus, if y is the output of the Purkinje cell and, in this simple model, taking into account the fact Purkinje cells are inhibitory

$$y = - \sum w_i x_i \quad (1)$$

where the x_i s are the activities in the parallel fibers and w_i is the strength of the synapse from the i th parallel fiber to the Purkinje cell. The idea in the Marr-Albus model is that the climbing fibre carries the error signal $d - y$ making the cerebellum an example of supervised learning.

Summary

The cerebellum is found at the back of the head and is thought to help the brain control movements, possibly by predicting the consequence of motor commands. Cerebellar damage causes cerebellar ataxia.

The cerebellum is specialised for rapid supervised learning: Massive numbers of granule cells generate parallel fibres that cross many Purkinje cells. Excitation from granule $\rightarrow$ Purkinje cells triggers Purkinje cell *simple spikes*, which are believed to approximate a prediction. Supervised, error-based learning is believed to occur when a climbing fibre from the inferior olive triggers a *complex spike* in a Purkinje cell.

References

- [1] Albus, JS. (1971) A theory of cerebellar function. *Mathematical Biosciences* 10: 25–61.
- [2] Gao J-H, Parsons LM, Bower JM, Xiong J, Li J and Fox PT (1996) Cerebellum implicated in sensory acquisition and discrimination rather than motor control. *Science* 272: 545–7.
- [3] Marr, D (1969) A theory of cerebellar cortex. *Journal of Physiology* 202: 437–70.